

VALVE SPRING DRAWING COMPARISON

Exhaust Spring A1770530600 vs Intake Spring A1770530500

Mercedes-Benz AMG M177 Engine · Scherdel GmbH · DIN 2098 Series

Wire Section	Free Length Δ	Seat Load Δ	Full-Lift Load	Avg. Work Rate
2.92 × 3.66 mm oval (shared)	Exhaust 47.0 mm Intake 46.1 mm (+0.9 mm)	Exhaust 270 N Intake 250 N (+8 %)	Both 620 N (shared)	Exhaust 35.0 N/mm Intake 37.0 N/mm

Engineering Drawings

Exhaust Valve Spring A1770530600

Technical specifications table:

Item	Description	Value	Unit
1	Wire diameter	2.92	mm
2	Inner diameter	3.66	mm
3	Outer diameter	4.58	mm
4	Free length	47.0	mm
5	Direction of cutting	Right	
6	Direction of cutting	Left	
7	Direction of cutting	Right	
8	Direction of cutting	Left	
9	Direction of cutting	Right	
10	Direction of cutting	Left	
11	Direction of cutting	Right	
12	Direction of cutting	Left	
13	Direction of cutting	Right	
14	Direction of cutting	Left	
15	Direction of cutting	Right	
16	Direction of cutting	Left	
17	Direction of cutting	Right	
18	Direction of cutting	Left	
19	Direction of cutting	Right	
20	Direction of cutting	Left	
21	Direction of cutting	Right	
22	Direction of cutting	Left	
23	Direction of cutting	Right	
24	Direction of cutting	Left	
25	Direction of cutting	Right	
26	Direction of cutting	Left	
27	Direction of cutting	Right	
28	Direction of cutting	Left	
29	Direction of cutting	Right	
30	Direction of cutting	Left	
31	Direction of cutting	Right	
32	Direction of cutting	Left	
33	Direction of cutting	Right	
34	Direction of cutting	Left	
35	Direction of cutting	Right	
36	Direction of cutting	Left	
37	Direction of cutting	Right	
38	Direction of cutting	Left	
39	Direction of cutting	Right	
40	Direction of cutting	Left	
41	Direction of cutting	Right	
42	Direction of cutting	Left	
43	Direction of cutting	Right	
44	Direction of cutting	Left	
45	Direction of cutting	Right	
46	Direction of cutting	Left	
47	Direction of cutting	Right	
48	Direction of cutting	Left	
49	Direction of cutting	Right	
50	Direction of cutting	Left	
51	Direction of cutting	Right	
52	Direction of cutting	Left	
53	Direction of cutting	Right	
54	Direction of cutting	Left	
55	Direction of cutting	Right	
56	Direction of cutting	Left	
57	Direction of cutting	Right	
58	Direction of cutting	Left	
59	Direction of cutting	Right	
60	Direction of cutting	Left	
61	Direction of cutting	Right	
62	Direction of cutting	Left	
63	Direction of cutting	Right	
64	Direction of cutting	Left	
65	Direction of cutting	Right	
66	Direction of cutting	Left	
67	Direction of cutting	Right	
68	Direction of cutting	Left	
69	Direction of cutting	Right	
70	Direction of cutting	Left	
71	Direction of cutting	Right	
72	Direction of cutting	Left	
73	Direction of cutting	Right	
74	Direction of cutting	Left	
75	Direction of cutting	Right	
76	Direction of cutting	Left	
77	Direction of cutting	Right	
78	Direction of cutting	Left	
79	Direction of cutting	Right	
80	Direction of cutting	Left	
81	Direction of cutting	Right	
82	Direction of cutting	Left	
83	Direction of cutting	Right	
84	Direction of cutting	Left	
85	Direction of cutting	Right	
86	Direction of cutting	Left	
87	Direction of cutting	Right	
88	Direction of cutting	Left	
89	Direction of cutting	Right	
90	Direction of cutting	Left	
91	Direction of cutting	Right	
92	Direction of cutting	Left	
93	Direction of cutting	Right	
94	Direction of cutting	Left	
95	Direction of cutting	Right	
96	Direction of cutting	Left	
97	Direction of cutting	Right	
98	Direction of cutting	Left	
99	Direction of cutting	Right	
100	Direction of cutting	Left	

Intake Valve Spring A1770530500

Technical specifications table:

Item	Description	Value	Unit
1	Wire diameter	2.92	mm
2	Inner diameter	3.66	mm
3	Outer diameter	4.58	mm
4	Free length	46.1	mm
5	Direction of cutting	Right	
6	Direction of cutting	Left	
7	Direction of cutting	Right	
8	Direction of cutting	Left	
9	Direction of cutting	Right	
10	Direction of cutting	Left	
11	Direction of cutting	Right	
12	Direction of cutting	Left	
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15	Direction of cutting	Right	
16	Direction of cutting	Left	
17	Direction of cutting	Right	
18	Direction of cutting	Left	
19	Direction of cutting	Right	
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21	Direction of cutting	Right	
22	Direction of cutting	Left	
23	Direction of cutting	Right	
24	Direction of cutting	Left	
25	Direction of cutting	Right	
26	Direction of cutting	Left	
27	Direction of cutting	Right	
28	Direction of cutting	Left	
29	Direction of cutting	Right	
30	Direction of cutting	Left	
31	Direction of cutting	Right	
32	Direction of cutting	Left	
33	Direction of cutting	Right	
34	Direction of cutting	Left	
35	Direction of cutting	Right	
36	Direction of cutting	Left	
37	Direction of cutting	Right	
38	Direction of cutting	Left	
39	Direction of cutting	Right	
40	Direction of cutting	Left	
41	Direction of cutting	Right	
42	Direction of cutting	Left	
43	Direction of cutting	Right	
44	Direction of cutting	Left	
45	Direction of cutting	Right	
46	Direction of cutting	Left	
47	Direction of cutting	Right	
48	Direction of cutting	Left	
49	Direction of cutting	Right	
50	Direction of cutting	Left	
51	Direction of cutting	Right	
52	Direction of cutting	Left	
53	Direction of cutting	Right	
54	Direction of cutting	Left	
55	Direction of cutting	Right	
56	Direction of cutting	Left	
57	Direction of cutting	Right	
58	Direction of cutting	Left	
59	Direction of cutting	Right	
60	Direction of cutting	Left	
61	Direction of cutting	Right	
62	Direction of cutting	Left	
63	Direction of cutting	Right	
64	Direction of cutting	Left	
65	Direction of cutting	Right	
66	Direction of cutting	Left	
67	Direction of cutting	Right	
68	Direction of cutting	Left	
69	Direction of cutting	Right	
70	Direction of cutting	Left	
71	Direction of cutting	Right	
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73	Direction of cutting	Right	
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75	Direction of cutting	Right	
76	Direction of cutting	Left	
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78	Direction of cutting	Left	
79	Direction of cutting	Right	
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81	Direction of cutting	Right	
82	Direction of cutting	Left	
83	Direction of cutting	Right	
84	Direction of cutting	Left	
85	Direction of cutting	Right	
86	Direction of cutting	Left	
87	Direction of cutting	Right	
88	Direction of cutting	Left	
89	Direction of cutting	Right	
90	Direction of cutting	Left	
91	Direction of cutting	Right	
92	Direction of cutting	Left	
93	Direction of cutting	Right	
94	Direction of cutting	Left	
95	Direction of cutting	Right	
96	Direction of cutting	Left	
97	Direction of cutting	Right	
98	Direction of cutting	Left	
99	Direction of cutting	Right	
100	Direction of cutting	Left	

A 177 053 06 00
Valve Spring – Exhaust (Auslassfeder)

A 177 053 05 00
Valve Spring – Intake (Einlassfeder)

Parameter Comparison Table

N o.	Symbol / Parameter	Unit	Exhaust A1770530600	Intake A1770530500	Δ / Remark
1	d — wire diameter (oval cross-section)	mm	2.92 × 3.66	2.92 × 3.66	Identical
2	Dio — inner dia. top (oben)	mm	12.00 ± 0.20	12.00 ± 0.20	Identical
3	Diu — inner dia. bottom (unten)	mm	15.90 ± 0.20	15.90 ± 0.20	Identical
4	Deo — outer dia. top	mm	19.32	19.32	Identical
5	Deu — outer dia. bottom	mm	23.22	23.22	Identical
6	# — spring index	—	5.32	5.32	Identical
7	Coiling direction	—	Right (RH)	Right (RH)	Identical
8	G — shear modulus	N/mm ²	79 500	79 500	Identical
9	L0 — free length	mm	47.0	46.1	Exhaust +0.9 mm
10	L1 — installed length	mm	36.10	36.10	Identical
11	L2 — max. working length	mm	26.10	26.10	Identical
12	Lc — solid length	mm	24.60 – 0.9	24.60 – 0.9	Identical
13	F1 — load at L1 (seat/installed)	N	270 ± 13	250 ± 12	Exhaust +20 N (+8 %)
14	F2 — load at L2 (full lift)	N	620 ± 27	620 ± 21	Same load; tighter tol. on intake
15	k_avg — avg. working rate (derived)	N/mm	35.0	37.0	Exhaust softer over stroke
16	τ_k1 — torsional stress at L1	N/mm ²	453	419	Exhaust +34 N/mm ² (+8 %)
17	τ_k2 — torsional stress at L2	N/mm ²	1 040	1 040	Identical
18	τ_kh — fatigue / stress range	N/mm ²	587	621	Intake +34 N/mm ² (+5.8 %)
19	nt — total coils	—	8.6	8.6	Identical
20	na — active coils	—	4.8 – 3.4	4.4 – 3.1	Different coil engagement range
21	e_min — min. coil spacing	mm	0.55	0.55	Identical
22	Spring shape	—	Beehive	Beehive	Identical
23	End treatment	—	Closed & ground 240°	Closed & ground 240°	Identical
24	Shot peening	—	Yes	Yes	Identical
25	Colour stripe 1	—	Violet RAL 4002	Violet RAL 4002	Identical — spring type marker
26	Colour stripe 2 (ID colour)	—	Brown RAL 8004	Yellow RAL 1006	DIFFERENT — position identifier
27	Surface protection	—	Wax	Wax	Identical
28	Heat treatment	—	Yes (DIN 7386.20)	Yes (DIN 7386.20)	Identical
29	Supplier / std.	—	Scherdel / DIN 2098	Scherdel / DIN 2098	Identical

N o.	Symbol / Parameter	Unit	Exhaust A1770530600	Intake A1770530500	Δ / Remark
30	Material	—	VD SiCrNiV SC	VD SiCrNiV SC	Identical (DIN 17 223)

★ Highlighted rows indicate parameters that differ between the two springs.

Correction Notice — Rev. 2 (2026-06-16)

An earlier draft of this report (Rev. 1) contained a transcription error in the intake spring fatigue stress range: τ_{kh} was incorrectly stated as 521 N/mm² instead of the drawing value of 621 N/mm². The error arose from misreading a digit in the reduced-resolution drawing thumbnail used for initial data extraction (the numeral '6' was read as '5'). The correct values satisfy the identity $\tau_{k1} + \tau_{kh} = \tau_{k2}$ for both springs:

Exhaust: 453 + 587 = 1 040 N/mm² ✓
Intake: 419 + 621 = 1 040 N/mm² ✓

The correction also updates the active-coil range (row 20): exhaust 4.8–3.4 (not 4.4–3.4) and intake 4.4–3.1 (not 4.4–3.4), read from high-resolution crops of the original TIF drawings. All analysis text and conclusions have been updated accordingly. The corrected values are highlighted in this table.

Engineering Analysis

1. Shared Design Basis

Both springs are manufactured by Scherdel GmbH to DIN 2098 / DIN 17 223 standards using **VD SiCrNiV SC** valve-spring steel wire with an **oval cross-section (2.92 × 3.66 mm)**. The beehive (Bienenkorb) profile, coiling direction (RH), spring index (5.32), coil count (nt = 8.6), end treatment (closed-and-ground, 240°), solid length (Lc = 24.60 mm), installed length (L1 = 36.10 mm), maximum working length (L2 = 26.10 mm), and full-lift force (F2 = 620 N) are all identical. Both springs receive the same shot-peening, wax surface protection, and heat treatment. This shared basis minimises part proliferation and tooling cost — the two parts diverge only where the valve dynamics demand it.

2. Free Length and Preload Difference

The sole geometric difference between the two springs is free length: **L0 = 47.0 mm (exhaust) vs 46.1 mm (intake)**, a delta of 0.9 mm. Because both springs are installed to the same assembly height (L1 = 36.10 mm), the 0.9 mm extra free length compresses the exhaust spring 0.9 mm further at installation, raising the **seat load (F1) from 250 N to 270 N — an 8 % increase**. The torsional stress at the installed length rises proportionally: 453 N/mm² (exhaust) vs 419 N/mm² (intake).

3. Derived Spring Rates Over the Working Stroke

Using the simple average-rate formula $k = \Delta F / \Delta L$ over the 10 mm working stroke (L1 – L2):

Exhaust: $k = (620 - 270) / (36.10 - 26.10) = 35.0 \text{ N/mm}$

Intake: $k = (620 - 250) / (36.10 - 26.10) = 37.0 \text{ N/mm}$

Both springs reach the same force (620 N) at full valve lift, but the exhaust spring starts from a higher seat load, so its effective spring rate over the working stroke is slightly **lower (35.0 vs 37.0 N/mm)**. The intake spring therefore has approximately **5.7 % higher dynamic stiffness** across the lift event. Note that the beehive geometry produces a naturally progressive rate; these figures represent mean values across the stroke.

4. Fatigue Stress Range and Durability

The fatigue stress range $\tau_{kh} = \tau_{k2} - \tau_{k1}$ is the primary HCF driver for valve springs (verified identity: $\tau_{k1} + \tau_{kh} = \tau_{k2} = 1\,040 \text{ N/mm}^2$ for both springs). Contrary to initial intuition, the **intake spring carries the higher stress range (621 vs 587 N/mm², +5.8 %)**. This is a direct arithmetic consequence of the lower seat load: because the intake spring starts each lift event at a lower torsional stress (419 vs 453 N/mm²) but reaches the identical peak (1 040 N/mm²), it must swing through a larger amplitude every cycle.

- **Exhaust spring:** higher mean stress (453 N/mm² at seat), smaller stress range (587 N/mm²). On a Goodman diagram the higher mean stress shifts the operating point toward the UTS intercept, demanding a material with good creep resistance at elevated temperature — met by VD SiCrNiV SC + shot-peening.
- **Intake spring:** lower mean stress (419 N/mm²), larger stress amplitude (621 N/mm²). The greater cyclic excursion is the governing fatigue driver; the lower mean stress partially offsets this on the Goodman line.

Both springs share the same peak stress at full lift ($\tau_{k2} = 1\,040 \text{ N/mm}^2$), confirming the valvetrain is dimensioned to the same maximum stress level; it is only the *starting point* of each load cycle that differs.

5. Load Tolerance and Assembly Quality

The full-lift tolerance on the exhaust spring is $\pm 27 \text{ N}$ vs $\pm 21 \text{ N}$ for the intake spring. A tighter tolerance on the intake spring may reflect closer attention to intake air-flow symmetry and cylinder-to-cylinder variation, whereas the exhaust spring relies more on the higher absolute preload level to guarantee closure. Both are well within the limits prescribed by DIN 2098 Part 2 (Gütevorschrift) and subject to statistical process control (Mitteldruckprüfen, DIN 6515 escalation level C, verified by Scherdel).

6. Colour Coding and Field Identification

Both springs share the first colour stripe (violet RAL 4002) indicating the spring *type* (beehive, oval wire, AMG series). The **second colour stripe uniquely identifies the position**:

- Exhaust: **brown (braun) RAL 8004**
- Intake: **yellow (gelb) RAL 1006**

This two-colour coding system prevents incorrect installation during engine assembly or service — the springs are externally identical in shape, and only the colour stripe distinguishes them without measurement.

Conclusions

- **Near-identical springs, one key difference:** The exhaust spring is the same part as the intake spring except for a 0.9 mm longer free length, which raises the seat load by 20 N (8 %). The intake spring, starting from a lower seat load but reaching the same peak force, carries a 5.8 % higher fatigue stress range (621 vs 587 N/mm²).
- **Same peak force at full lift (620 N):** Both springs are designed to deliver the same maximum force at full valve lift. This ensures the valvetrain cam/follower system, designed to a single peak force level, can be applied to both banks without modification.
- **Higher exhaust preload is physically justified:** The additional seat load on the exhaust spring (270 vs 250 N) compensates for combustion back-pressure on the exhaust valve, preventing valve float or delayed closure under high-RPM or high-load conditions.
- **Intake spring has slightly higher working stiffness:** Paradoxically, the softer (lower-preload) intake spring shows a 5.7 % higher average working rate (37.0 vs 35.0 N/mm) over the valve lift event, because it must traverse the same force range from a lower starting point.
- **Fatigue dimensioning: intake swing is larger:** Both springs share the same peak torsional stress at full lift (1 040 N/mm²). Because the intake spring starts each cycle from a lower seat stress (419 vs 453 N/mm²), it traverses a larger stress amplitude (621 vs 587 N/mm²). The exhaust spring compensates with a higher mean stress level — both trade-offs are addressed by the VD SiCrNiV SC material and shot-peening process.
- **Colour coding is critical for correct assembly:** The brown vs yellow second stripe is the only visual distinction between the springs. Swapping them would install too-high preload on intake and too-low on exhaust — causing potential intake-side over-stress and exhaust-side valve float.
- **No changes to manufacturing process or supplier:** Both parts share supplier (Scherdel), material specification, end treatment, surface protection, and quality standard. The free-length difference is the sole production distinguisher, minimising supply-chain complexity.

Summary: The exhaust and intake valve springs of the M177 engine are nearly identical beehive springs sharing all geometry and manufacturing parameters. The single engineering difference — a 0.9 mm longer free length on the exhaust spring — deliberately raises the seat load to resist exhaust back-pressure, while both springs converge to the same 620 N force at maximum valve lift. Correct installation is assured through a distinctive colour-stripe identification system (brown = exhaust, yellow = intake).

Document prepared automatically from engineering drawings A1770530600-3 and A1770530500-4. Derived calculations use nominal drawing values. All specifications subject to Scherdel DIN 2098 Part 2 quality protocol.